Day 20: Complete Chi-Square Workflow: From Table Data to P-Value Percentage

Advanced Data Analytics Notes

1. Definition of Chi-Square Test

English: The Chi-Square Test is a statistical hypothesis test used to determine whether there is a significant association or relationship between two categorical variables, comparing observed frequencies with expected frequencies based on a null hypothesis.

Hindi: *Chi-Square Test ek statistical hypothesis test hai jiska use yeh pata karne ke liye kiya jata hai ki kya do categorical variables ke beech koi mazboot rishta ya association hai, jisme hum dukan ke asli data (Observed) aur umeed kiye gaye data (Expected) ke bhatkaw ki tulna karte hain.*

English Explanation: This test evaluates whether the deviations in your data are simply a result of random chance or if they represent a genuine underlying business pattern.

Hindi Explanation: *Yeh test check karta hai ki aapke data mein jo badlav ya bhatkaw dikh raha hai woh sirf kismat ka ek ittefaq hai ya uske peeche koi asli business trend chipa hua hai.*

2. Part 1: Generating Chi-Square & Degree of Freedom From Table Data

English: Before computing probability percentages, we must generate the raw Chi-Square score and Degree of Freedom using a real cross-tabulation table layout.

Hindi: *Probability percentage nikalne se pehle hume ek real business table ka use karke raw Chi-Square score aur Degree of Freedom calculate karna hoga.*

Business Problem (English): A jewelry store tracks customer preference across two categories to see if Customer Type impacts the product purchased. Let's analyze the Observed Data table.

Business Problem (Hindi): Ek jewelry store do alag customer types (Tourist vs Local) aur unki product choice (Rings vs Chains) ka data track karta hai taaki check kiya ja sake ki kya inke beech koi relation hai.

Observed Data Table (O):

Customer Type	Rings	Chains	Row Total
Tourist	50	150	200
Local	150	50	200
Column Total	200	200	**Grand Total (N) = 400**

Expected Data Table Calculation (E):

English: Expected value for each cell is calculated as: (Row Total × Column Total) / Grand Total.

Hindi: *Har ek cell ke liye Expected value ka formula hota hai: (Row Total x Column Total) / Grand Total.*

- For all cells: $(200 \times 200) / 400 = 100$

Calculating Chi-Square Value (χ^2):

English: The formula is $\sum [(O - E)^2 / E]$ for all cells combined.

Hindi: *Chi-Square nikalne ka standard formula hai:* $\sum [(O - E)^2 / E]$. *Isko har cell ke liye solve karenge:*

- Tourist-Rings: $(50 - 100)^2 / 100 = 2500 / 100 = 25$
- Tourist-Chains: $(150 - 100)^2 / 100 = 2500 / 100 = 25$
- Local-Rings: $(150 - 100)^2 / 100 = 2500 / 100 = 25$
- Local-Chains: $(50 - 100)^2 / 100 = 2500 / 100 = 25$

$$\text{Total } \chi^2 = 25 + 25 + 25 + 25 = 100$$

Calculating Degree of Freedom (df):

English: Formula: (Rows - 1) × (Columns - 1).

Hindi: *Degree of Freedom ka formula hota hai: (Rows - 1) × (Columns - 1).*

$$df = (2 - 1) \times (2 - 1) = 1 \times 1 = 1$$

3. Part 2: Universal Power Series Formula Definition

English: Now we use the continuous power series model to transform the raw score and df values into an exact probability edge.

Hindi: *Ab hum is continuous power series formula ka use karke hamare calculated score aur df ko exact percentage mein badalna seekhenge.*

$$\text{P-Value} = [e^{-\chi^2/2} \cdot (\chi^2/2)^{(df/2) - 1}] / \Gamma(df/2)$$

4. Step-by-Step Numerical Case Study (For df=5, χ^2=24)

Case Parameter Setup: Let's take the complex condition where **df = 5** and **χ^2 = 24** to compute final percentage manually.

Step 1: Simplifying Parameters

- $\chi^2 / 2 = 24 / 2 = 12$
- $Power = (df / 2) - 1 = (5 / 2) - 1 = 1.5$
- $\Gamma(df / 2) = \Gamma(2.5) = 1.3293$

Step 2: Processing Component Values

1. **Auditor Part:** $e^{-12} = 0.000006144$
2. **Risk Manager Part:** $12^{1.5} = 41.569$
3. **Judge Part:** 1.3293

Step 3: Compounding & Final Percent Transformation

$$P\text{-}Value = (0.000006144 \times 41.569) / 1.3293 = 0.000192$$

$$Percentage = 0.000192 \times 100 = 0.0192\%$$

5. Final Statistical Interpretation & Decision Matrix

English: Since the exact P-value (0.0192%) is far below the alpha threshold of 5.0%, the null hypothesis is rejected, establishing a verified structural pattern.

Hindi Faisla: *Kyunki exact P-value (0.0192%) hamari maximum threshold limit (5%) se bohot choti hai, isiliye hum Null Hypothesis ko reject karte hain aur taye karte hain ki inke beech 100% solid relation hai.*

Metric Parameter	Calculated Value	Standard Threshold	Statistical Conclusion
Degrees of Freedom (df)	5	-	**Null Hypothesis Rejected. Framework relationship is mathematically verified.**
Chi-Square Score (χ^2)	24.0	Critical Line: 11.07	
Exact P-Value Percentage	**0.0192%**	Alpha Limit: 5.0%	